

Paper from counselor notes

1. Pallotta et al. (2013) - *Vessel Pattern Knowledge Discovery from AIS Data* (Entropy)

AIS Data Characteristics

- AIS is mandatory for vessels ≥ 300 gross tonnage on international voyages; fishing vessels $> 15\text{m}$ in EU waters also required
- Transmission rate is variable: every 2 minutes at anchor, up to every 2 seconds when maneuvering/high speed; every 5 minutes for static/voyage data
- Class B devices (common on smaller fishing vessels) transmit at lower rates than Class A, important limitation for fishing vessel datasets
- SOG (speed over ground) field in AIS messages is explicitly flagged as unreliable for detecting stationary events; recommend computing speed from $\Delta\text{pos}/\Delta t$ instead
- Gaps in AIS tracks caused by: terrestrial coverage gaps, intermittent AIS signal, satellite revisit time, or intentional transponder shutdown (distinct)

Segmentation / Gap Rule

- Vessel labeled "lost" after not being observed for time τ , where τ is defined as a multiple of the maximum AIS refresh rate in the area. i.e., the threshold is **area-dependent, not fixed**
- Status transitions used as segmentation events: "sailing" $\rightarrow$ "stationary" $\rightarrow$ "lost" $\rightarrow$ "sailing"

Feature Implications

- State vector includes position (x, y), COG, SOG, and timestamp (basically what we have?)
- Velocity components derived by combining SOG and COG: $\dot{x} = \text{SOG} \cdot \sin(\text{COG})$, $\dot{y} = \text{SOG} \cdot \cos(\text{COG})$ possibly more physically meaningful than using raw SOG alone

- Local directional distributions (heading distributions within a spatial neighborhood) used for route classification, suggests turning rate and heading variability are key discriminators

Traffic Entropy Concept (maybe important for strong labels in nordic ocean data)

- More constrained/regulated traffic → lower entropy → anomaly detection works better
- In open ocean fishing grounds, traffic is less constrained → higher entropy → harder to classify behavior purely from movement patterns. Relevant to the thesis's scope limitations.

Anomaly Detection as Related Problem

- Deviations from learned normal routes flagged as anomalies using posterior probability threshold
- Sliding time window used for online anomaly detection, avoids needing complete trajectories
- The double U-turn example is directly analogous to fishing behavior (repeated course reversals).

For the thesis

- good for: gap-based segmentation rationale, SOG unreliability, AIS transmission rate variability, Class B device limitations
- The trajectory representation formalism and velocity derivation from SOG+COG is directly usable

2. Wang & Pun (2022) - *Robust Semisupervised Land-Use Classification Using Remote Sensing Data With Weak Labels* (IEEE Access)

Core Problem Framing, relevant

- Three challenges addressed: class imbalance, label noise, image uncertainty. Maps almost exactly onto our challenges: class imbalance (fishing is minority), weak GFW labels (label noise), AIS gaps and spoofing (data uncertainty)

- Labels divided into two sets: $X(c)$ = accurate/strong labels, $X(b)$ = weak/boundary labels, this is conceptually identical to our kystverket experiment

Weak Label Definition

- Weak labels defined as those in boundary/uncertain regions, not necessarily wrong, but less certain
- Key insight: training on weak labels with standard cross-entropy can cause the classifier to incorrectly merge classes
- Label noise modeled as: $\tilde{y} = y^* + \epsilon$, where ϵ is label noise

Semisupervised Strategy

- Core idea: use small accurately-labeled set to define class boundaries, then use weakly-labeled data through unsupervised loss rather than supervised loss
- Two-component loss function:
 - LICE (Improved Cross-Entropy with TF-IDF weighting) for accurate-label data
 - LNuN (Nuclear norm maximization) for weak-label data
 - Combined: $L = LICE + \lambda \cdot LNuN$
- λ is a hyperparameter controlling contribution of weak supervision that needs tuning

Handling Class Imbalance - TF-IDF Weighting

- Standard cross-entropy biases toward majority class (non-fishing dominating AIS data)
- TF-IDF-inspired weighting: classes that appear frequently in one image but rarely across the dataset get higher weight - minority classes upweighted
- For us: fishing segments are the minority class; standard training will bias toward non-fishing
- Simpler alternatives: `class_weight='balanced'` in sklearn, or SMOTE oversampling

Nuclear Norm for Weak Labels

- Rather than minimizing cross-entropy on weak labels (which amplifies noise), maximize nuclear norm of prediction matrix
- Nuclear norm $\approx$ convex relaxation of matrix rank - maximizing it encourages the model to predict diverse classes rather than collapsing to majority class
- Prevents the diversity collapse shown in Figure 4 (could attach here im just lazy) classifier trained on weak labels alone tends to merge minority classes
- This is a deep learning technique (requires prediction matrix from neural network) - not directly applicable to random forest/logistic regression, but relevant if moving to neural network models

"Unknown" Class

- Artificial class added to handle data with atypical/unrepresentative features
- Weakly-labeled uncertain samples get absorbed into "unknown" rather than forcing misclassification
- Prevents overfitting to noisy boundary samples
- For us: could conceptually treat ambiguous AIS segments (e.g., vessel drifting in fishing ground but not actively fishing) as an "unknown" class rather than forcing binary labels

Experimental Design (most relevant?)

- Paper uses both a weak-label test set and a separately curated accurate-label test set for evaluation
- This is experiment 2 and 3 (test on GFW weak labels vs test on Kystverket strong labels)
- Key finding: model trained and tested on weak labels can look deceptively good; testing on accurate labels reveals true performance gap
- Their semisupervised approach trained on weak labels approached performance of supervised model trained on accurate labels, so hybrid approach has real value

Key Results

- Semisupervised ICE+NuN outperformed supervised CE by ~10% mIoU on minority classes
- Faster convergence with nuclear norm (removes confusing boundary features early in training)
- Performance gains most pronounced for minority/rare classes (fishing class)
- Tested across multiple CNN backbones (U-Net, FCN, DeepLabV3+), NuN improvement held across all

For us

- Strong methodological justification for the three-experiment design
 - Confirms that testing on strong labels after training on weak labels is the correct way to reveal label quality impact
 - TF-IDF weighting idea translatable to simpler class weighting for random forest
 - Nuclear norm idea is future work / extension if moving to neural models
 - Cite for: weak label noise formulation, class imbalance handling strategies, motivation for the strong-label test set, semisupervised learning logic
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3. Pipanmekaporn & Kamonsantiroj (2020) - *A Deep Learning Approach for Fishing Vessel Classification from VMS Trajectories Using RNNs* (Springer IHET)

Problem

- Classification of fishing vessel *types* (not fishing vs non-fishing) from VMS trajectory data
- Three gear types: trawl, purse seine, falling net, each has distinct movement signatures
- VMS data is similar to AIS but purpose-built for fisheries monitoring; lower transmission frequency typically

Preprocessing

- Density-based clustering (DBSCAN-like) used to eliminate meaningless data points and identify coherent movement groups
- Local trajectories generated from these groups, segmentation but using spatial density rather than time gaps
- This is probably something to highlight in the segmentation choice

Feature Engineering

- Feature vector computed per identified trajectory group (not per raw AIS point)
- Changing trend of angle (COG) and speed (SOG) selected as key input parameters, directly validates our choices
- Features fed into RNN as sequence rather than aggregated statistics, big difference from us

Models Compared

- Multiple RNN architectures tested
- Bidirectional GRU achieved best performance: **89.74% accuracy**
- GRU (Gated Recurrent Unit) is lighter than LSTM but handles sequential dependencies well
- Bidirectional processes sequence both forward and backward, capturing context from both directions

Key Finding

- Sequence-based models (RNN/GRU) outperform feature-aggregation approaches for trajectory classification when temporal order matters
- This is the core tradeoff with using feature-based (random forest/logistic regression) rather than sequence-based models - current limitation (for report)

What this means for the report

- Validates that COG trend (turning rate) and SOG trend (speed variability) are the right features

- The bidirectional GRU result is a strong benchmark (**89.74%**) to reference when discussing what performance might be achievable with a more sophisticated model
- The DBSCAN preprocessing step is an alternative to time-gap segmentation and can mention in discussion
- Directly citable as: prior work showing fishing vessel classification from trajectory data is feasible with RNNs; contrast with the thesis's simpler feature-based approach and explain why simplicity is appropriate for the stated research question (maybe hard to argue if results aren't close?)

4. Wang et al. (2020) - *Learning from Weak-Label Data: A Deep Forest Expedition* (6092Article, AAAI-20)

Core Problem

Weak-label learning addresses the case where each instance has multiple true labels but only a subset is observed, the rest are silently missing, not confirmed negative. This is a precise formal analogue to our GFW label construction: segments with `fishing_hours = 0` may simply be undetected, not confirmed non-fishing.

LCForest

Proposes a cascade forest (layer-by-layer deep forest) with an embedded **label complement structure**. In each layer, random forests trained on one cross-validation fold predict labels for the held-out fold. If predictions are consistently positive for a label that is currently negative in the weak matrix, that label is promoted (complemented) for the next layer. A **complement flag mechanism** using the Tlce label frequency estimator stops the cascade when the estimated positive label count has been sufficiently recovered, preventing over-complementing without a ground-truth validation set.

Key hyperparameters: T=10 layers, K=5 folds, threshold $\theta=0.4$ (below 0.5, intentionally biased toward predicting positives to counteract false negatives).

Why Relevant

- Formalizes exactly the label noise structure we face: $\tilde{y} = y^* + \epsilon$ where missing positives are the main problem

- Demonstrates that tree ensemble methods (the same family as our random forest) can handle weak supervision robustly, and outperform neural networks on small/medium datasets under label noise
- The threshold-biasing-toward-positives idea ($\theta < 0.5$) is directly transferable: lowering the decision threshold on our RF's probability output could recover fishing segments that GFW missed
- Label complement is a possible future extension if we had multiple behavior labels (fishing type), we could iteratively refine them this way

Experiments & Results

Tested on Yeast (14 labels), TMC-2007 aviation text (22 labels), Medical NLP (45 labels) with IL ratios 20–50%. LCForest achieves best Micro-F1 and Hamming Loss across all settings and is robust to increasing incompleteness. The competitor WELL is briefly better on Macro-F1 at 20% IL but deteriorates rapidly at higher ratios.

Key point: **robustness to unknown noise level** is the critical property, not peak performance at a specific known ratio, directly applicable since we cannot know GFW's true error rate.

5. Ji et al. (2023) - *Multi-label Classification with Weak Labels by Learning Label Correlation and Label Regularization* (Multilabel_classification, Applied Intelligence)

Core Problem

Most MLL methods address only one weakness: missing labels, noisy labels, or class imbalance. Real data has all three simultaneously. GFW labels are noisy/incomplete, and fishing is the minority class, both problems compound each other.

Method ML-INC

Decomposes the observed label matrix Y into a low-rank incomplete ground-truth component and a sparse noisy component using **low-rank + sparse decomposition** (SVD-based). A **label confidence matrix** is learned by exploiting high-order label correlations and label consistency. Assigning real-valued confidence rather than binary labels to uncertain assignments. A **label regularization matrix** counteracts class imbalance by reweighting the label space.

A sparse L1 constraint on the feature mapping matrix simultaneously performs feature selection. Solved via ADMM.

Why Relevant

- The low-rank label correlation idea is the theoretical underpinning of why aggregated trajectory statistics (mean speed, std speed, turning rate) are informative: vessel behavior classes occupy a low-dimensional manifold in feature space
- The simultaneous treatment of noise + imbalance is the strongest justification for why handling them independently (e.g., `class_weight='balanced'` for imbalance, then ignoring label noise) is suboptimal
- Even for our binary setting: the confidence matrix idea translates to soft probabilistic labels for borderline GFW segments, rather than forcing hard 0/1 labels that propagate noise directly into training

Experiments & Results

16 datasets (4 small, 6 medium, 6 large). ML-INC achieves best performance in 48% of cases and top-2 in 88%. Strongest gains on *Scene* and *Corel5k*, small, severely imbalanced datasets analogous to our. Friedman test confirms statistically significant superiority on Coverage, Ranking Loss, Average Precision, and AUC. The ADMM convergence is verified on Medical, Genbase, and Yeast datasets.

6. Huang et al. (2018) - *FVID: Fishing Vessel Type Identification Based on VMS Trajectories* (J. Ocean Univ. China)

probably most relevant if we wanna get a really good result, specifically feature enumeration/selection

Core Problem

VMS trajectories can identify that a vessel is fishing but not *what kind* of fishing, yet fishing activity recognition requires vessel type as a prerequisite, because speed/direction thresholds differ by gear type. 1,350 of 11,381 vessels in the East China Sea dataset are unregistered with unknown types. Goal: classify all vessels into 9 types using trajectory features only.

Data

49.2 million VMS records (5-minute resolution, China Beidou Satellite), March 2017, East China Sea. 9 types: shrimp trawl (2761), otter trawl (1581), pair trawl (1364), gill net (2093), canvas stow net (727), crab cage (292), square net (271), light seine (135), transportation (807). Heavy class imbalance, similar to our dataset.

Method FVID Pipeline

1. Preprocessing: drop null GPS records; compute average speed/direction between consecutive records rather than instantaneous (more stable given sensor noise and vessel motion)
2. Feature enumeration: 61 features. Speed divided into 4 bands (0-0.5, 0.5-2.5, 2.5-5.5, 5.5-12 knots); features capture histogram counts per band, mean/median/std per band, fishing proportion by time-of-day window (8 windows), and spatial mean/median lat/lon
3. Feature selection: variance threshold to remove low-variance features → XGBoost Gain-based wrapper selection; feature count varies by vessel type (Otter Trawl optimal at 23 features)
4. Classifier: XGBoost with One-vs-Rest (OVR) multi-class strategy
5. SMOTE oversampling for the 3 minority types (T6, T7, T8)

Results

95.42% 4-fold CV accuracy on registered vessels. The speed distribution analysis (Fig. 2) is crucial as it shows that different gear types have distinctly different speed profiles across the 4 bands. Shrimp and otter trawl concentrate in 2.5–5.5 knots during fishing; gill net has a bimodal distribution at 0.5–2.5 and 5.5–12; transportation is almost entirely 5.5–12 knots.

Why Directly Relevant

- The speed band visualization (Fig. 2) is the empirical foundation for our feature choices: speed mean, std, and the fishing-speed heuristic (2–5 knots). Cite this
- Their 61-feature space is good ideas for expansion of our features (time-of-day proportions, spatial position) could be a future work direction

- XGBoost vs our RF: both are tree ensemble methods; 95.42% accuracy is strong
- SMOTE for minority classes: we should apply this or equivalent class weighting; this paper validates the approach
- The 5-minute VMS resolution vs AIS's variable rate shows that fixed-interval data makes feature engineering cleaner

7. Xing et al. (2023) - *Fishing Vessel Behavior Identification Based on AIS Data: East China Sea* (JMSE, MDPI)

Core Problem

AIS data is noisier and lower-resolution than VMS but far more available. Goal: classify 3 fishing behavior types (purse seine, gill net, trawl operation) directly from AIS trajectories for fishery regulation. Downstream motivation: combating IUU fishing, which captures 11–26 million tons annually (~15% of global fish consumption).

Method

- **Cubic spline interpolation** to standardize the irregularly sampled AIS trajectory (fills gaps, normalizes to a uniform time grid)
- **Geohash encoding**: divides the sea area into a spatial grid; maps trajectory points to grid cells and extracts a discrete sequence of cell IDs, converts spatial trajectories into token sequences
- **CBOW (Continuous Bag of Words)** applied to the grid cell sequences, treating cells as words and learning trajectory embeddings from co-occurrence context
- **LightGBM** classifier trained on the trajectory embeddings, with Bayesian hyperparameter optimization

Results

Training accuracy 0.925; F1-score improvement +1.8% over XGBoost, +1.2% over CatBoost. Bayesian optimization outperforms manual tuning. The combined trajectory vectorization + LightGBM pipeline substantially outperforms raw statistical features fed into standard gradient boosting.

Why Relevant

- Directly comparable domain (AIS, fishing behavior classification, East China Sea, multiclass). A ~0.93 F1-score is achievable with this approach. ambitious ceiling for us
 - **Cubic spline interpolation** is the methodological solution to the AIS irregular sampling problem that the current time constraint rule does.
 - Geohash + CBOW is a sequence-aware representation that captures spatial context our feature-based approach discards entirely, a natural future direction
 - Bayesian hyperparameter optimization is a systematic alternative to our manual parameter setting - relevant to model comparison methodology discussion
 - This paper cites FVID (Huang 2018) directly, confirming they are in the same methodological lineage as our work
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8. Cheng et al. (2025) - *Attention-enhanced and Integrated Deep Learning for Fishing Vessel Classification Based on Multiple Features* (Scientific Reports)

Core Problem

Most existing fishing vessel classification models handle 2–4 vessel types and focus on either static or dynamic features in isolation. Vessel type labels are often self-reported, mislabeled, or missing, creating incomplete monitoring analogous to our GFW weak label problem. Goal: 5-class classification (gillnetter, hook-and-liner, trawler, fish carrier, stow net vessel) from AIS data using a multi-feature deep learning ensemble.

Data

Chinese offshore AIS, National Data Center for Distant-water Fisheries. Training: 2018–2021 (~6.25 million trajectory records, 12,485 labeled vessels). Independent test: 2022 (1,000 vessels, 200 per class, balanced). Preprocessing: remove duplicates, remove speed > 12 knots (outlier threshold), segment by day, remove trajectories < 6 hours or < 160 AIS messages. SMOTE applied to balance training classes - good justifications for us doing this stuff

Feature Engineering, 58 features (reduced to 46 by selection)

Three types combined:

- *Geometric* (4D): vessel length, width, area, length/width ratio, static registry-level information
- *Static* (42D): for each of longitude, latitude, speed (overall + 3 speed segments: 0-2, 2-8, 8-12 knots), heading - compute mean, upper quartile, median, lower quartile, STD, COE (coefficient of variation). Speed segment 2-8 knots (fishing speed range) statistics are the top-ranked features by both RF Gini and XGBoost Gain
- *Dynamic* (12D): statistical summaries of Δ time, Δ longitude, Δ latitude between consecutive AIS points, can captures local movement dynamics that aggregated statistics miss

Feature selection: correlation heatmap to identify collinear pairs → RF Gini + XGBoost Gain importance ranking → wrapper forward selection removes the bottom 12 features → 46-dimensional final vector.

Most important feature: COE of speed in the 2-8 knot segment (F33), validates that variability within the fishing speed range is the single most discriminating feature. This directly supports our use of speed standard deviation.

Model BiLSTM-CNN-Attention

9-layer ensemble: 2× bidirectional LSTM layers (capture global sequence dependencies forward and backward) → reshape → 2× Conv2D layers (extract local pattern features) → MaxPooling → Flatten → Attention layer (weight-assignment to important features) → fully connected softmax output. Trained with Adam, categorical cross-entropy, batch size 64, 100 epochs.

Results

Model	Accuracy	F1-Score
SVM	74.1%	74.1%
RF	90.2%	90.3%
XGBoost	88.3%	88.4%
BiLSTM	89.5%	89.7%
BiGRU	89.7%	89.8%

Model	Accuracy	F1-Score
BiLSTM-CNN-Attention	91.9%	91.4%

Independent test, 10-fold CV stable. CNN performs well in CV but poorly on independent test, instability flags it as unsuitable for this task. RF is the best-performing single non-sequential model (90.2%), closely trailing the full deep learning ensemble.

Common misclassification: gillnetters and stow net vessels confused with trawlers, the largest class. Model bias toward majority class under class imbalance is visible even after SMOTE.

Why Big

- **RF at 90.2% F1 on a 5-class problem** is a critical benchmark. Our binary task is simpler, so our RF's performance should be interpretable relative to this ceiling. If we fall significantly below 90%, something is wrong with our feature set, preprocessing, or label quality, not inherently with RF as a method
- The finding that COE of speed in 2-8 knots is the top feature validates mean speed and speed std as our core features. Our turning rate maps conceptually onto their $\Delta\text{lat}/\Delta\text{lon}$ dynamic features
- Speed segment statistics (0-2, 2-8, 8-12 knots) are a natural expansion of our feature space, easy to implement and directly motivated by this paper
- SMOTE application and the resulting performance gives us a concrete justification for applying it to our class-imbalanced fishing/non-fishing data
- The confusion matrix analysis showing majority-class bias even after SMOTE is directly relevant to our discussion of precision/recall tradeoffs and why F1 and PR-AUC are good metrics

Lookup

Theme	Papers
Weak/noisy/missing labels formal treatment	Wang (2020), Ji (2023), Wang & Pun (2022)*
Class imbalance handling (SMOTE, weighting)	Ji (2023), Huang (2018), Cheng (2025)

Theme	Papers
Speed as primary discriminating feature	Huang (2018) Fig. 2, Cheng (2025) F33, Pallotta (2013)*
AIS irregular sampling & preprocessing	Xing (2023) interpolation, Pallotta (2013)* gap rule
RF/XGBoost as strong baseline for AIS classification	Huang (2018) 95.4%, Cheng (2025) RF 90.2%
Deep learning vs feature-based tradeoff	Cheng (2025) BiLSTM-CNN-Attention, Pipanmekaporn (2020)* BiGRU
Evaluation under label uncertainty (dual test set)	Wang & Pun (2022)*, our experiment design
Feature importance analysis methodology	Cheng (2025) RF Gini + XGBoost Gain, Huang (2018) Wrapper